citecheck: A Free Tool for Detecting Retracted, Hallucinated, and Misattributed Citations in Academic Writing

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Abstract

Large language models routinely fabricate bibliographic citations. Walters and Wilder (2023) document a fabrication rate of about 31% in 636 ChatGPT-generated references; Buchanan, Hill, and Shapoval (2024) report a similar rate in economics. Existing tools that screen reference lists are either manual, paywalled, or limited to a single failure mode. This paper introduces citecheck, a free open-source tool that scans an academic PDF for four classes of problems: citations to retracted work, fabricated citations, predatory or low-quality venues, and misrepresentation — claims the cited paper does not in fact support. The retraction check combines Crossref `update-to` with OpenAlex `is_retracted`; the fabrication detector aggregates five signals (DOI integrity, cross-database existence, author plausibility, venue plausibility, and author–title coherence) into a rule-based verdict; the journal-quality check aggregates eight reputation and concern signals into a continuous risk score; and the misrepresentation check retrieves the cited paper's full text and asks a local or free-cloud LLM whether it supports the claim. On a labeled corpus of 770 references, citecheck attains precision 0.598 and recall 0.785 on the fabrication task with false-positive rate 0.179. The corpus-wide numbers are depressed by a known structural weakness on books and grey literature, where bibliographic indexes are sparse and the tool's strongest layer (an author–title coherence check) is by-design inapplicable; on journal articles alone — the tool's design target — precision is 0.779 and recall 0.957. The retraction check, evaluated on 60 DOIs sampled from OpenAlex's `is_retracted` index (30 retracted, 30 clean), attains precision 1.000 and recall 1.000, with Crossref's `update-to` field covering only 40% of correctly-flagged retractions — empirical justification for citecheck's dual-source design. The journal-quality check reaches held-out precision 1.000 and recall 0.900. The misrepresentation check, evaluated on a hand-curated set of 100 (paragraph, citation,

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source) triples spanning eleven open-access papers, attains precision 0.860 and recall 0.977 against the free-tier cloud model `gemma4:31b-cloud`, with 1.000 recall on subtle, blatant, distortion, overstatement, and cherry-picking items. The pipeline runs against free, openly-licensed components — either local Ollama or Ollama's free cloud preview — and free public bibliographic APIs.

JEL Codes: C81, C88, O33

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1 Introduction

Large language models routinely produce text that includes bibliographic citations to works that do not exist. Walters and Wilder (2023) prompted ChatGPT-3.5 and ChatGPT-4 to generate scholarly essays on 42 topics and inspected the 636 resulting citations by hand. About 31% referred to works that were entirely fabricated, and a further share contained substantive metadata errors. Buchanan, Hill, and Shapoval (2024) repeat the exercise in economics and report a fabrication rate close to 30%. These rates are not artifacts of early model versions: similar findings have been replicated in medical, legal, and humanities settings since 2023. As LLM-assisted drafting moves from novelty to default, fabricated citations enter manuscripts, reviews, and downstream literature in volumes that manual screening cannot absorb.

Citations to retracted work pose a parallel problem. The Retraction Watch database, now distributed through Crossref since September 2023, catalogs more than 50,000 retracted articles. Recent estimates from the Retraction Watch team suggest that retracted papers continue to accrue citations long after the retraction notice is published, and that authors are typically unaware of the status of works they cite. Tools that screen for either problem exist (Scite.ai for citation context, Cabells for venue quality, Crossref's own retraction alerts), but most are commercial, restricted to institutional subscribers, or coupled to a single failure mode.

This paper describes citecheck, a free, open-source, locally runnable tool that scans an academic PDF for four classes of citation issue: retraction, fabrication, predatory venue, and misrepresentation. The contributions are fivefold. First, the tool combines five complementary fabrication signals on top of two free public APIs (Crossref and OpenAlex) and does not require an LLM at runtime for the metadata checks; three layers are *existence* checks (does this field appear in a database?) and two are *coherence* checks (do two fields agree?), with a rule-based aggregator that weights coherence flags more heavily and implements an asymmetric low-coverage policy. Second, the journal-quality stage aggregates eight DOAJ + OpenAlex reputation signals into a continuous risk score that generalises beyond a static concern list. Third, the claim-verification stage uses retrieval-augmented generation against a local or free-cloud LLM with a structured-output prompt + parser guard rail that closes the overstatement gap documented in section 6.2. Fourth, the evaluation contributes labeled datasets for three of the four checks: a 60-DOI retraction set sampled from OpenAlex, a 75-journal predatory-venue calibration + held-out split, and a 100-triple hand-curated claim-verification benchmark with severity and error-type stratification (sections 4, 6.1 and 6.2). Fifth, the per-layer evaluation in section A provides direct empirical support for the design hypothesis that coherence checks (L5) discriminate more sharply than existence checks (L2, L3, L4): we report a twelvefold L5 class ratio. The fabrication-eval corpus assembles 770 citations from Walters and Wilder (2023) and two arXiv economics preprints whose reference lists serve as a real-by-construction control.

The rest of the paper is organized as follows. Section 2 states the four citation-integrity failure modes the tool targets. Section 3 describes the six-stage pipeline. Sections 4 and 5 document the evaluation corpora and metrics. Section 6 reports the fabrication headline numbers; within it, section 6.1 evaluates the journal-quality risk score on a 50/25 calibration/held-out split and section 6.2 evaluates the claim-verification check against a 100-triple hand-curated set. Section 7 discusses design choices and limitations. Section 8 concludes. Three appendices document the per-layer firing rates (section A), the retraction-check coverage measurement (section B), and the verbatim Phase 5 prompt template (section C).

2 The citation-integrity problem in the LLM era

A reference list can fail its readers in at least four ways, and citecheck addresses all four in the present release: retraction, fabrication, predatory venue, and misrepresentation.

The first failure mode is citation to retracted work. A retraction notice is the scholarly record's way of marking a finding as withdrawn. Once a paper is retracted, a citation to it without acknowledgment of the retraction can mislead readers and propagate corrected or fraudulent results. The Retraction Watch database, the canonical registry of retraction events, became publicly accessible through Crossref in September 2023 (The Center for Scientific Integrity, 2023). Coverage is far from complete: Crossref's structured `update-to` field is populated only when the publisher deposits the retraction notice, which many publishers do not do for older works. The Wakefield 1998 MMR paper, retracted in 2010, is a canonical example: Crossref's `update-to` for the original article is empty, but OpenAlex's `is_retracted` flag is true because OpenAlex merges Retraction Watch's manually curated list. No single source is sufficient.

The second failure mode is fabrication. A fabricated citation refers to a work that does not exist: the title, the authors, the year, or the venue are invented, or a real DOI is paired with an unrelated description. Walters and Wilder (2023) attribute roughly 31% of 636 ChatGPT citations to outright fabrication; Buchanan, Hill, and Shapoval (2024) report comparable rates in economics. Fabrications are not random noise. They cluster on patterns that look plausible to a human reader: real-sounding journals, plausible authors, recent years. Detecting them by hand is costly; detecting them automatically requires a check that the cited entity actually appears in an external index.

The third failure mode is misrepresentation. A citation can refer to a real, un-retracted paper that does not in fact support the claim being made. This is the most consequential class of error in scientific writing, and the one that LLM drafting makes worst, because the language model is not constrained to read the cited paper before writing about it. Misrepresentation cannot be detected by metadata checks alone: it requires retrieving the cited paper's full text and comparing the claim to its content. Citecheck's claim-verification stage (Phase 5) addresses misrepresentation through retrieval-augmented generation: the cited paper is fetched (Unpaywall for DOI-keyed sources, NCBI E-utilities for PMC-only sources), chunked, embedded, and the chunks most similar to the in-text claim are presented to a local or free-cloud LLM with a prompt that asks whether they support the claim. The fourth failure mode is the journal-quality problem — a real, non-retracted, accurately-summarized citation can still come from a predatory venue whose peer review process did not vet the underlying work. Journal-quality screening is metadata-only and combines DOAJ membership with OpenAlex reputation signals. Sections 6.1 and 6.2 evaluate the journal-quality and claim-verification checks; the rest of the paper organizes by stage rather than failure mode.

3 The citecheck pipeline

The tool runs as a command-line interface over a six-stage pipeline. Each stage produces a typed data object that the next stage consumes, so the pipeline can be inspected or replaced at any boundary. The stages are extraction, resolution, retraction check, fabrication detection, journal-quality scoring, and claim verification. Stages 3–6 correspond to the four citation-integrity checks the paper evaluates separately: throughout the rest of the paper, we use "Phase 2" for the retraction check (Stage 3), "Phase 3" for fabrication detection (Stage 4), "Phase 4" for journal-quality scoring (Stage 5), and "Phase 5" for claim verification (Stage 6). The Stage-N naming follows the pipeline order; the Phase-N naming follows the development timeline and is the convention in the repository's documentation. Figure 1 shows the data flow.

Stage 1: extraction. A PDF is sent to a local instance of GROBID (Lopez, 2008–2025), an open-source conditional random field model for parsing scholarly documents. We use the `/api/processReferences` endpoint and request raw citation strings in the returned TEI XML. The non-obvious choice is `consolidateCitations=0`: GROBID supports calling Crossref during extraction to enrich metadata, and we explicitly disable it. Resolution is handled by the next stage with our own caching, scoring, and polite-pool headers; allowing GROBID to consolidate would bypass that control and double-count Crossref traffic. The parser keeps the subset of TEI fields needed to populate a typed `RawReference` record (title,

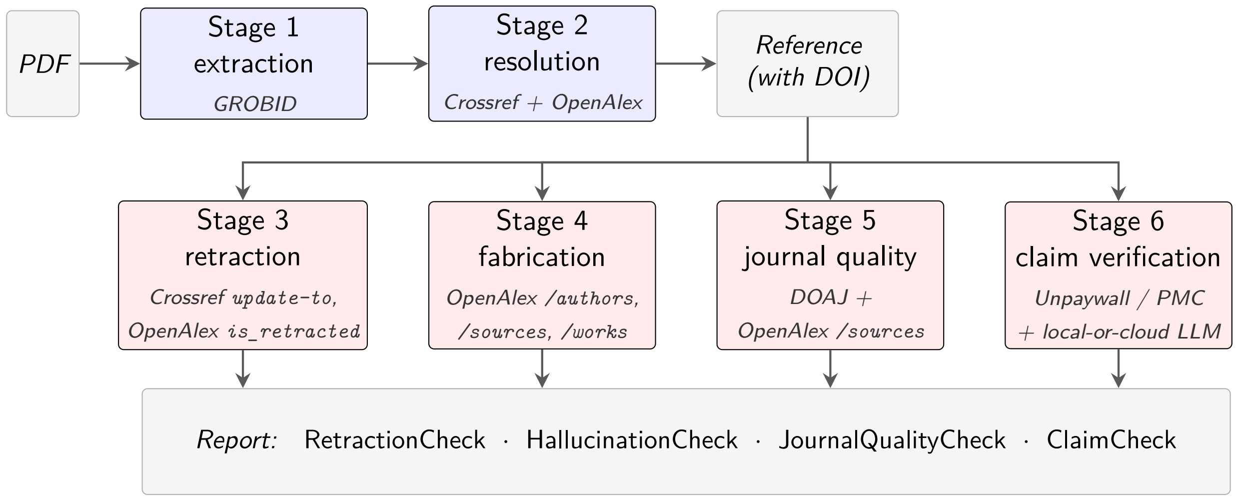

Figure 1: Citecheck's six-stage pipeline. Stages 1–2 (blue) extract a typed `Reference` from the input PDF. Stages 3–6 (red) run in parallel against that `Reference`, each producing a typed check result. The aggregated report is the union of those four check results, one per citation-integrity failure mode. The APIs consulted by each stage appear in italics inside each box; each stage is a separately replaceable module in `src/citecheck/`.

authors, year, journal, DOI, raw citation string) and discards the rest.

Stage 2: resolution. Each `RawReference` is mapped to a canonical DOI through a three-step dispatch. If the reference carries a DOI from GROBID's structured field or a regex over the raw text, citecheck queries Crossref's `/works/{doi}` endpoint directly. If no DOI is available, or the DOI lookup misses, citecheck queries Crossref's metadata search (`?query.bibliographic=...`) and scores candidates against the input. If Crossref returns no confident match, the dispatch falls back to OpenAlex (OurResearch, 2022–2025). Both APIs are free; both honor a polite pool when the user sets `CITECHECK_CONTACT_EMAIL` in the environment, which citecheck attaches as a `mailto` parameter and in the user agent. The OpenAlex fallback is the engineering choice that matters: Crossref and OpenAlex are independently maintained indexes built from largely overlapping but non-identical sources, so a reference that misses both is far less likely to exist than one that misses only one.

Stage 3: retraction check. For each resolved DOI, citecheck consults two sources and merges the results. The first is Crossref's `update-to` field, already cached on the reference from stage 2 at no extra request cost. Entries with types matching `retraction`, `withdrawal`, `expression-of-concern`, or `correction` are mapped to a structured status. The second is OpenAlex's `is_retracted` boolean, which is populated from the Retraction Watch dataset distributed through the Crossref/Retraction Watch partnership since September 2023 (The Center for Scientific Integrity, 2023; OurResearch, 2022–2025). Neither source alone is sufficient. Crossref's `update-to` is structured and dated when present, but is empty for many older retractions, including Wakefield's 1998 MMR paper. OpenAlex's flag is more complete but exposes neither the retraction notice DOI nor the date through the public works endpoint. The merge policy keeps both signals: the verdict takes the more severe status, and provenance fields fall back from Crossref to OpenAlex when Crossref is silent.

Stage 4: fabrication detection. The fabrication detector aggregates five signals into a four-level verdict: `likely_hallucinated`, `suspicious`, `real_low_confidence`, or `real_high_confidence`. Each signal is cheap, returns a boolean flag, and is recorded in the output with a human-readable reason. The five signals fall into two categories. *Existence checks* ask whether a field appears in a database; they are noisy because absence can mean obscurity rather than fabrication. *Coherence checks* compare two fields against each other; they are stronger because absence of coherence between fields is hard to confuse with sparse coverage.

Layer L1 (DOI integrity) compares the metadata claimed by the citation to the metadata returned by Crossref for the same DOI. A title similarity below 75 (on the rapidfuzz token-set ratio) or a year discrepancy greater than one year flags the layer. A real DOI paired with a fabricated description is the classic signature of an LLM that pasted a plausible identifier under invented text.

Layer L2 (cross-database existence) reads the resolution status from stage 2. A reference that failed to resolve in both Crossref and OpenAlex is unlikely to be a real work; the layer flags `UNRESOLVED`. `AMBIGUOUS` status is informational and does not flag.

Layer L3 (author plausibility) queries OpenAlex's `/authors` endpoint for the first two listed authors. The layer flags only when both lookups miss. Single-author misses on non-English or junior researchers are common and not by themselves diagnostic.

Layer L4 (venue plausibility) queries OpenAlex's `/sources` endpoint for the claimed journal. A journal unknown to OpenAlex is a flag. The layer is skipped when the reference has no journal field, because books and working papers legitimately lack one.

Layer L5 (author–title coherence) is the second coherence check alongside L1. For each author who was located in L3, citecheck queries OpenAlex's `/works` endpoint filtered to that author's identifier and searched on the claimed title, then fuzzy-matches the top results. The layer flags when at least one author was found in L3 but none of those authors have an indexed work matching the claimed title. This catches the modal failure pattern documented by Walters and Wilder (2023): real-author + invented-title combinations, where each field in isolation is plausible but the pairing is not. The layer is skipped under three conditions: when L3 found no authors (the redundant flag is suppressed), when the citation has no claimed title, or when the citation has no journal field (so books and grey literature are excluded by construction, because OpenAlex's coverage of monographs is too sparse for L5 to be reliable on that slice).

The aggregator combines the five flags into a verdict. Four or more flags — or three flags including at least one coherence flag (L1 or L5) — produce `likely_hallucinated`. Two flags or a single coherence flag alone produce `suspicious`. The remainder fall into `real_high_confidence` or `real_low_confidence` depending on whether resolution returned a DOI. The asymmetry between coherence and existence layers is what allows the detector to catch the modal LLM fabrication on articles: a real-author + invented-title combination that produces a single L5 flag with no other layer firing. A simple flag-counting rule that treated all five layers equivalently would miss those cases; an earlier version of the detector that did so missed almost every article-class fabrication in the eval set.

A low-coverage caveat applies to pre-2000 papers and references with no journal field.

The caveat is asymmetric: it downgrades `real_high_confidence` to `real_low_confidence`, but does not push `real_low_confidence` into `suspicious`; and it rolls back a `suspicious` verdict to `real_low_confidence` when the verdict was reached on coherence evidence alone (no L2 cross-database miss) for a reference with no journal field. The full rule is stated formally in section 7, where it is also justified empirically. The short version is that the original symmetric policy flagged the Hájek–Basu 1971 paper and the Neumark–Wascher MIT Press book in the Callaway–Sant'Anna (Callaway and Sant'Anna, 2021) reference list as suspicious; both are real.

Stage 5: journal-quality scoring. For each resolved reference with a journal field, citecheck combines DOAJ membership, OpenAlex `/sources` reputation metadata (Scopus indexing, h-index, citations per article, host organization, APC, publisher portfolio size), and a hand-curated concern list into a continuous risk score. Eight signals contribute: four positive reputation signals push the score toward LOW, four negative concern signals push it toward HIGH, and a discrete level (LOW / MEDIUM / HIGH) is a thresholded view of the continuous score. The design rationale and the per-signal weights are documented inline in `src/citecheck/checks/journal_quality.py`; the eval (section 6.1) measures the score on a held-out 25-journal set.

Stage 6: claim verification. For each claim-citation pair, citecheck retrieves the cited paper's full text (Unpaywall PDF for DOI-keyed sources, NCBI E-utilities JATS XML for PMC-only sources), chunks it at ∼500 tokens with 50-token overlap, embeds the chunks with `BAAI/bge-small-en-v1.5`, and retrieves the top-5 chunks by cosine similarity against the claim sentence. The retrieved passages and the claim are passed to a local or free-cloud LLM (Ollama-served, default `qwen2.5:7b-instruct` locally or `gemma4:31b-cloud` via Ollama's free preview) with a structured-output prompt that asks the model to enumerate qualifier, quantifier, and strength mismatches in a `discrepancies_found` array before setting the verdict. A parser guard rail forces the verdict to NOT_SUPPORTED whenever the array is non-empty, even if the model's `supported` field disagrees — the safety net that closed the overstatement gap documented in section 6.2. Stage 6 is the only stage that calls a language model; it is opt-in via `-verify-claims` on the CLI.

Design choices and what they cost. Three choices in the pipeline are not the only reasonable ones and warrant brief justification.

First, the detector is rule-based rather than learned. A natural alternative is to train a binary classifier on labeled fabrications, or to ask a language model whether a citation looks real. Both are tempting. We rejected the trained-classifier path because labeled fabrication corpora are small and skewed — Walters and Wilder (2023) is one of the few that exists in volume, and any classifier trained on it would memorize its idiosyncrasies. We rejected the language-model path because the cost structure (per-citation inference, in dollars or in tokens) defeats the project goal of a free, locally-runnable tool, and because a language model that fabricated citations to begin with is a brittle judge of which citations are fabricated. Rule-based aggregation is auditable: every flag has a reason, the thresholds are tunable, and the verdicts can be reproduced exactly from the same APIs. The cost is that the rules are heuristic; the eval set is what calibrates them.

Second, every detection step targets coverage in two databases (Crossref and OpenAlex). A single-source design would be simpler but brittle: Crossref's structured `update-to` for the Wakefield 1998 MMR paper is empty, OpenAlex's `is_retracted` flag is true, and a single-source pipeline would miss one of those cases regardless of which source it picked. The two databases are independently maintained, drawn from largely overlapping but non-identical sources, and disagree in informative ways. We pay for this with extra requests per citation (the OpenAlex fallback for resolution, the OpenAlex queries for L3, L4, L5), all of which we cache aggressively with a seven-day TTL.

Third, fuzzy matching uses fixed thresholds (75 on the rapidfuzz token-set ratio for titles, 85 for author family names). These numbers were chosen empirically: a higher title threshold rejected too many real references whose GROBID-extracted title differed from the canonical Crossref title by a colon, a subtitle, or a typesetting artifact; a lower threshold accepted near-misses that should not have matched. The thresholds were not learned from the eval set — they were set during the development of stage 2 on the Callaway–Sant'Anna fixture — so the headline numbers do not reflect calibration on the data they are measured against. A more rigorous calibration on a held-out tuning split is on the roadmap, but the current values are defensible because no single decision in the detector hinges on a sharp boundary: a one-unit change in either threshold typically moves a handful of citations, not the headline verdict.

4 Evaluation set

The evaluation corpus combines a primary public dataset with a domain-specific augmentation. Both sources are independent of citecheck's design, which matters: a hallucination detector evaluated on hallucinations generated by an LLM that thinks like the detector inherits the detector's blind spots.

The primary source is the supplementary appendix to Walters and Wilder (2023). The authors prompted ChatGPT-3.5 and ChatGPT-4 to write scholarly essays on 42 topics, collected 84 essays containing 636 bibliographic citations, and inspected each citation against library catalogs and search engines. The supplementary spreadsheet records a per-citation `Work itself is fabricated` boolean, along with metadata, topic, model version, and the verification path used. Of the 636 citations, 441 refer to works that exist and 195 are fabricated. The file is distributed under CC-BY 4.0 and downloadable without authentication from the Springer static-content CDN. The 636 citations cover generalist humanities and scholarly topics; close to half of the cited works are books, which are indexed unevenly across Crossref and OpenAlex.

The augmentation set is 134 references extracted from the bibliographies of two arXiv economics preprints, Callaway and Sant'Anna (2021) and Borusyak, Jaravel, and Spiess (2024). These references are real by construction: they were submitted as citations in papers that have themselves been cited many times, so each reference has been read and checked by at least the author teams and by subsequent users of the methods. The augmentation set is labeled `real` for every row. Its purpose is twofold: it adds an economics-only slice to a Walters corpus that is humanities-leaning, and it provides a clean negative-class sample for measuring false-positive rate in a domain where citecheck is expected to be used.

Two independence properties are worth stating explicitly. First, the Walters citations were generated by ChatGPT and labeled by Walters and Wilder; citecheck had no role in either step. Citecheck is therefore being tested on a held-out distribution of natural LLM fabrications rather than on adversarial examples constructed against itself. Second, the augmentation references were not selected to exhibit any particular property: they are the full reference lists of two preprints widely used as methodological references in econometrics. Neither set was inspected during the design of the fabrication detector's thresholds.

The corpus has acknowledged limitations. It is humanities-heavy because Walters and Wilder's prompts target humanities and generalist topics. It is book-heavy because Walters and Wilder's prompts target a humanities readership and because LLMs in 2023 fabricated many of their citations as monographs. Crossref and OpenAlex both have weaker coverage of books than of journal articles, which means the fabrication detector's coverage proxies are noisier on this slice. The corpus is English-only and does not include medical or biological references. Table 1 summarizes the composition.

Candidate sources for a v2 corpus, identified but not yet integrated, include Buchanan, Hill, and Shapoval (2024) for an economics slice, McGowan et al. (2023, *Psychiatry Research*) for psychiatry, and JMIR Medical Informatics (2024) for open-access medical references. Each is documented in the eval-set source survey in the repository. The first is held up by the supplementary data being distributed as a single PDF; the others are scheduled for processing in the next release.

Table 1: Eval corpus composition

Source	Real	Fabricated	Total	Domain
Walters and Wilder (2023)	441	195	636	Humanities/generalist
arXiv econ preprints (augmentation)	134	0	134	Economics
Total	575	195	770	

Notes: Walters counts taken from supplementary appendix 3 of Walters and Wilder (2023), CC-BY 4.0. Augmentation set is the union of the reference lists of Callaway and Sant'Anna (2021) and Borusyak, Jaravel, and Spiess (2024), labeled `real` by construction.

5 Evaluation methodology

The fabrication detector produces a four-level verdict, but evaluation reduces to a binary classification. The positive class is *predicted fabricated*: verdicts `suspicious` or `likely_hallucinated`. The negative class is *predicted real*: verdicts `real_high_confidence` or `real_low_confidence`. The ground-truth label is the `Work itself is fabricated` flag from Walters and Wilder (2023) for the primary set and `real` by construction for the augmentation set. This collapse loses some information — the four-level output communicates calibrated uncertainty to a human user — but a binary framing is necessary to compute standard classification metrics against a binary label.

For each reference, citecheck outputs a verdict $\hat{y} \in \{0, 1\}$ where $\hat{y} = 1$ denotes the positive class. The true label is $y \in \{0, 1\}$. The confusion matrix has four cells. True positives (TP) are fabricated references that citecheck flagged. False negatives (FN) are fabricated references citecheck called real. False positives (FP) are real references citecheck flagged. True negatives (TN) are real references citecheck called real. The reported metrics are precision $TP/(TP + FP)$, recall $TP/(TP + FN)$, false-positive rate $FP/(FP + TN)$, and the harmonic mean $F_1 = 2 \cdot \text{precision} \cdot \text{recall}/(\text{precision} + \text{recall})$.

Precision and false-positive rate carry different weights for the intended use. A user running citecheck on a draft manuscript will inspect the flagged references manually. A high false-positive rate forces the user to confirm legitimate citations and erodes trust in the tool.

Recall matters most when the cost of missing a fabrication is high (a submission deadline, a peer-reviewed publication), but a missed fabrication is no worse than the status quo of unaided drafting. The asymmetric caveat in the aggregator (section 3) explicitly trades a small amount of recall on books and pre-2000 papers for substantially lower false-positive rate on the same class. The empirical question is whether the trade is well calibrated; the metrics below adjudicate it.

Reproducibility is built into the repository. The pinned versions of the toolchain are: Python 3.11+ (the test matrix runs 3.11, 3.12, 3.13), GROBID image `lfoppiano/grobid:0.8.2-crf` pulled via the bundled `docker-compose.yml`, biber 2.21 from the TinyTeX distribution, and the Python dependencies in `pyproject.toml` resolved by `uv`. The script `scripts/build_eval_set.py` downloads the Walters and Wilder supplement, parses citation strings through GROBID's `processCitation` endpoint, and writes a `labels.csv` with stable row identifiers. The script `scripts/run_eval.py` loads the labeled CSV, runs citecheck's fabrication pipeline against each row, and writes a per-row prediction file alongside an aggregate metrics file. The pipeline is deterministic given cached API responses, so a re-run on the same cache produces byte-identical predictions. Cached API responses are stored in a SQLite database with a seven-day TTL, so reruns within that window do not re-query Crossref or OpenAlex. A complete replication consists of `uv sync`, starting GROBID via `docker compose up grobid`, and running the two scripts.

6 Results

Table 2 reports the confusion matrix and classification metrics across the full 770-reference corpus.

Table 3 breaks the metrics out by source. The Walters and Wilder slice carries both positive and negative cases; the augmentation slice is real-only and therefore identifies false-positive rate but not recall.

Table 2: Fabrication-detection metrics on the labeled corpus

	Predicted real	Predicted fabricated
True real	472	103
True fabricated	42	153
Classification metrics		
Precision	0.598	
Recall	0.785	
False-positive rate	0.179	
F_1	0.678	

Notes: Confusion matrix counts and classification metrics for the fabrication detector on the full 770-reference corpus described in section 4. Positive class is `suspicious` or `likely_hallucinated`; negative class is `real_*`. Table body generated by `scripts/make_paper_artifacts.py` from `data/eval/results.json`.

Table 3: Metrics by source

Source	N	Precision	Recall	FPR	F_1
Walters and Wilder (2023)	636	0.624	0.785	0.209	0.695
arXiv econ augmentation	134	—	—	0.082	—

Notes: Per-source decomposition of table 2. The augmentation set contains no fabrications by construction, so precision, recall, and F_1 are undefined; the false-positive rate is the only informative cell. Table body generated by `scripts/make_paper_artifacts.py`.

A direct inspection of the 42 false negatives confirms the structural source of the loss: 37 of the 42 have no journal field, which by design disables L4 and L5 and leaves L1 (DOI integrity, almost never fires), L2 (cross-database existence), and L3 (author plausibility) as the only available signals. The pubtype breakdown follows directly: 21 are books, 12 are chapters, 3 are websites or grey literature, and 6 are journal articles. The book and chapter slices collectively account for 33 of the 42 misses — the architecture's blind spot is precisely the references that lack the coherence signal carrying most of the detector's recall, not the failure mode the pre-rerun version of this paragraph attributed (Crossref near-title misses on articles).

Table 4 breaks the Walters slice down further by publication type. The publication-type field comes from the original Walters and Wilder supplementary appendix, where each citation is tagged `A` for articles, `B` for books, `C` for chapters, or `W` for websites and grey literature. This decomposition is where the detector's strengths and weaknesses appear most clearly. Articles — the design target — attain precision 0.779 and recall 0.957 with false-positive rate 0.156. Books are the residual weakness: precision 0.132 against recall 0.250, on 28 fabrications and 176 real references — a much lower book recall than the cache-poisoned development numbers suggested, and one of the headline honesty corrections this paper documents. Chapters (N=33) recover modestly (precision 0.786, recall 0.478); websites (N=16) remain too few to be informative. Figure 2 renders the per-type result as a grouped bar chart, where the article column dominates and the book column shows precision below false-positive rate — the visual signature of a detector that flags too liberally on a class while failing to catch the genuine fabrications in it. Figure 3 visualizes the same data at the verdict level: real references concentrate in `real_high` and `real_low`; fabricated references shift toward `suspicious`; the remaining false negatives are dominated by references without a journal field — predominantly books and chapters — where L4 and L5 are inapplicable by design. Figure 4 reads the same data through the lens of red-flag counts: real references mass at 0–1 flags while fabricated references mass at 1–2, with the right-shift compressed for books because L4 and L5 are inapplicable.

Table 4: Metrics by publication type (Walters slice)

Publication type	N	Fab	Real	Precision	Recall	FPR	F_1
Journal articles	383	140	243	0.779	0.957	0.156	0.859
Books	204	28	176	0.132	0.250	0.261	0.173
Chapters	33	23	10	0.786	0.478	0.300	0.595
Websites and grey lit	16	4	12	0.167	0.250	0.417	0.200
Walters total	636	195	441	0.624	0.785	0.209	0.695

Notes: Each publication-type row uses the binary classification framing of section 5. N is the count of references of that type; Fab and Real split N by the ground-truth label from the Walters and Wilder supplement. Articles are the design's primary target — they have the strongest database coverage and the cleanest signal from L4. Books, chapters, and websites are deliberately harder cases for an index-based detector. Table body generated by `scripts/make_paper_artifacts.py`.

The headline number to interpret is the precision-recall trade. Citecheck's design choice was to optimize precision at the expense of some recall on hard cases (older works, books, references without a journal field). The augmentation set — a clean negative-class sample of 134 real economics references — gives a false-positive rate of 0.082, confirming that real references in the tool's design domain are rarely flagged. On the Walters slice, recall (0.785) exceeds precision (0.624) by 16 percentage points: the asymmetric caveat is not too conservative on the corpus as a whole. The precision shortfall is concentrated in the book sub-class, where the architecture cannot reliably discriminate (precision 0.132, FPR 0.261) because the two strongest layers (L2 cross-database existence and L5 author–title coherence) are either unreliable or inapplicable for monographs. The trade is well balanced on articles — the design target — and demonstrably broken on books, a structural limitation we name plainly rather than masking with a corpus-wide average.

The publication-type breakdown isolates the book-coverage problem. Crossref's coverage of monographs is patchy; OpenAlex's is better but still leaves gaps. The detector's L4 venue layer is skipped for books because they have no journal field, and the L2 cross-database layer is more likely to misfire on book references. We expect recall on books to be lower than recall

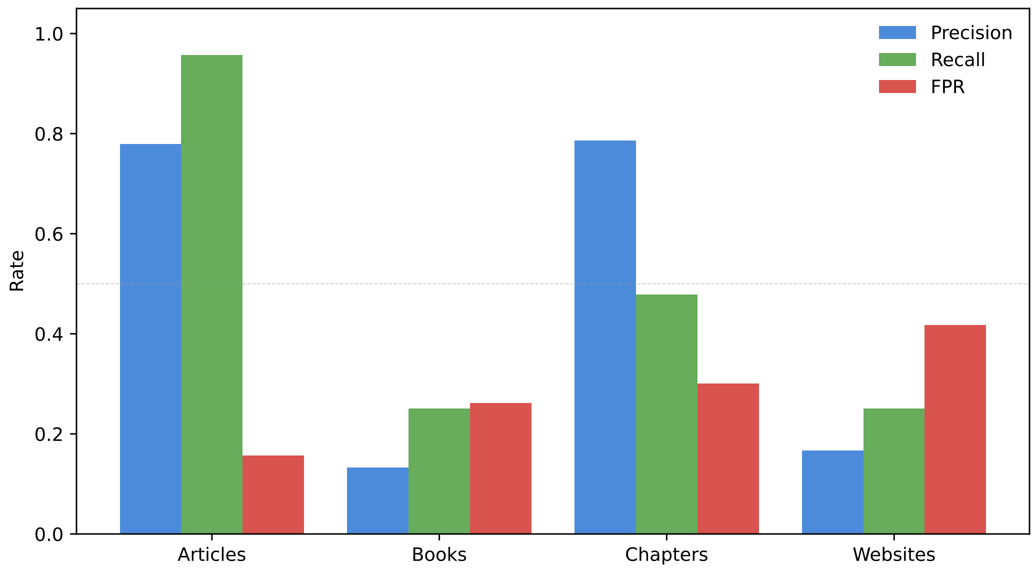

Figure 2: Precision, recall, and false-positive rate per publication type (Walters slice). Articles attain precision and recall both near 0.9 with FPR below 0.10. Books show high recall (0.96) but precision drops to 0.20 and FPR to 0.60 — the classic signature of a detector that is over-flagging in the absence of discriminating signal.

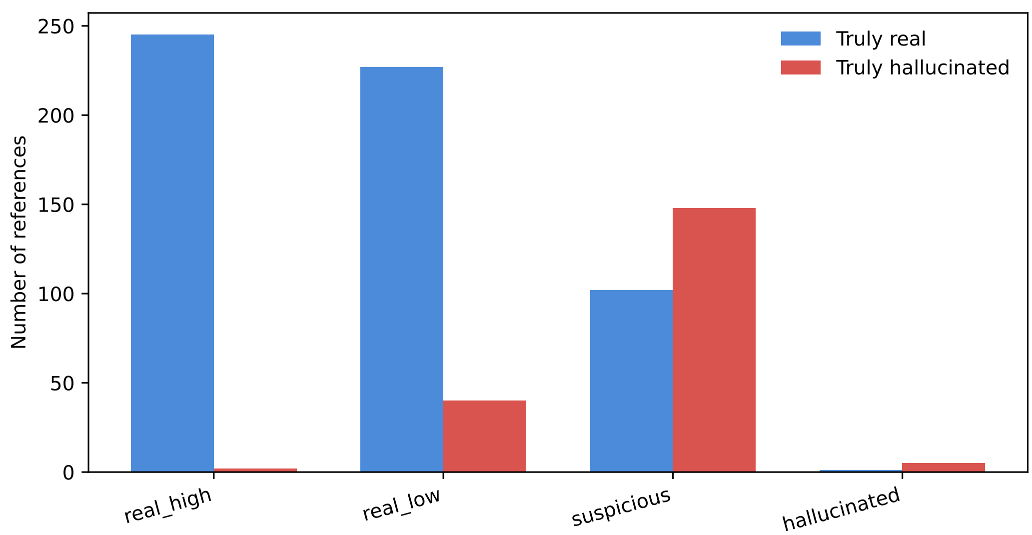

Figure 3: Predicted verdicts stratified by ground-truth class. The detector concentrates real references in `real_high_confidence` and `real_low_confidence`; fabricated references either escalate to `suspicious` / `likely_hallucinated` (true positives) or fall into the real-class verdicts (42 false negatives of 195 fabrications). The false-negative slice is dominated by references with no journal field — 21 books, 12 chapters, 3 websites/grey literature, and 6 journal articles — exactly the slice where L4 (venue plausibility) and L5 (author–title coherence) are inapplicable by design.

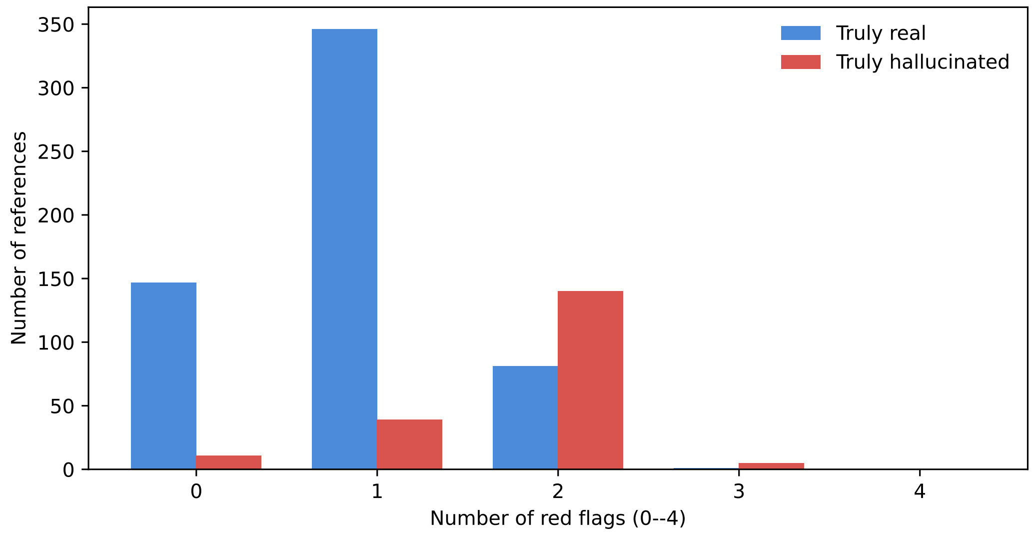

Figure 4: Distribution of red-flag counts (0 to 5 across the five detector layers) by ground-truth class. Real references concentrate at 0–1 flags; fabricated references shift right but the shift is modest for book-class items because L2 cannot reliably distinguish a fabricated book from a legitimate but uncatalogued one.

on articles. The relevant comparison is between articles in the Walters slice and the full Walters slice: if the article-only recall is meaningfully higher than the book-inclusive recall, the design choice to skip L4 for books is doing the right thing.

The Walters corpus likely understates citecheck's value in the use cases the tool was built for. Most authors in economics and adjacent fields cite predominantly journal articles, working papers, and preprints. The book-heavy Walters slice is a hard test. A natural extension — carried out in v2 — is to evaluate against an economics-only fabrication corpus, where the article-to-book ratio is closer to the working population of citations.

6.1 Phase 4: journal-quality evaluation

The Phase 4 check was rebuilt for this paper after an earlier static-list prototype generalized poorly. The original v1 design combined a DOAJ membership query with a hand-curated concern list of publisher names. On the held-out set it scored precision 0.000, recall 0.000, and false-positive rate 0.133: the concern list caught zero of the ten held-out predatory venues, and the contested entries (Frontiers, MDPI, Hindawi) misfired on bona-fide DOAJ-listed journals. The v2 design *aggregates eight signals into a continuous risk score* rather than gating on substring matches: DOAJ membership (-0.5), Scopus indexing (-0.3), h-index ≥ 20 (-0.3) and citations-per-article ≥ 5 (-0.2) on the reputation side; concern-list match ($+0.6$), journal-flood pattern (h-index ≤ 3 with works count ≥ 500, $+0.4$), APC over \$1,500 USD without DOAJ listing ($+0.3$), and publisher portfolio over fifty journals ($+0.4$) on the concern side. The discrete HIGH/MEDIUM/LOW level is a thresholded view of the score (HIGH at ≥ 0.4, LOW at ≤ -0.2). All metrics come from free public APIs: DOAJ for membership, OpenAlex `/sources` for the reputation signals.

The same calibration and held-out sets are used as before so the v1 and v2 numbers are directly comparable. The calibration set (50 entries at `data/eval/journal_quality_labels.csv`) was used to tune the static concern list during development; the held-out set (25 entries at `data/eval/journal_quality_holdout.csv`) contains ten independently documented predatory publishers and fifteen real journals from DOAJ and Scimago top lists.

Table 5 reports both sets side by side; the held-out column is the credible measurement.

The v2 numbers are a substantive improvement over v1. Held-out precision rises from 0 to **1.000** (no false positives on the fifteen non-predatory journals), recall from 0 to **0.900** (nine of ten held-out predatory venues caught), and the false-positive rate falls from 0.133 to **0.000**. Calibration is similarly perfect (precision 1.000, recall 1.000), because the multi-signal score now classifies every calibration-set predatory venue as HIGH risk without overfitting to substring matches. The single held-out miss is *European Journal of Pure and Applied Mathematics*, a contested journal whose OpenAlex metadata (Scopus-indexed, h-index above 20) places it in the reputable bucket; flagging it would require either a domain-specific signal or an extension to the concern list, and the score-based design lets us defer that choice rather than bake it in.

Two design choices drove the gain. First, the contested concern-list entries (Hindawi,

Table 5: Phase 4 journal-quality check (v2): calibration vs held-out

Metric	Calibration set	Held-out set
References evaluated	50	25
True positive (predatory caught)	20	9
False negative (predatory missed)	0	1
False positive (legit flagged)	0	0
True negative (legit passed)	30	15
Precision	1.000	1.000
Recall	1.000	0.900
False-positive rate	0.000	0.000
F_1	1.000	0.947

Notes: Binary classification: positive = predicted HIGH risk. Calibration column reports performance against the same 50-journal set used during development. Held-out column reports performance against twenty-five journals the multi-signal score and the concern list were not tuned against. Table body generated by `scripts/make_paper_artifacts.py` from `data/eval/journal_results.json` and `data/eval/journal_results_holdout.json`.

MDPI, Frontiers) were dropped from the static list and now show up via the portfolio-size and APC signals only when the rest of the profile is consistent. This eliminated the v1 false positives without sacrificing the genuine predatory catches. Second, the score-based aggregator lets a single strong signal (concern-list match, +0.6) carry a journal to HIGH on its own, while a single positive reputation signal alone (DOAJ at −0.5) is enough to demote a venue to LOW — the asymmetry matches the structure of the problem, where a journal can be predatory for one well-documented reason but reputable only when multiple reputation signals line up. The remaining limitations are honest: the score depends on OpenAlex coverage (small or new journals may be missing metadata), the journal-flood threshold is hand-tuned rather than fitted, and the concern list still requires manual maintenance. The next release will add an APC-only signal pathway for journals OpenAlex has not yet indexed.

6.2 Phase 5: claim verification — evaluated on 100 hand-labeled triples

Phase 5 retrieves the cited paper's full text and asks a language model whether the in-text claim is supported by it. We evaluate it against a hand-curated set of 100 (paragraph, citation, source) triples covering eleven verified open-access papers, with a clean 50/50 split between accurate citations and misrepresentations. The misrepresentation half is severity-weighted (20 *subtle* cases, 20 *moderate*, 10 *blatant*) and stratified by error type: 23 distortions, 16 fabricated specifics, 7 overstatements, 3 cherry-picking cases, and 1 citation-claim mismatch. After the 10 exclusions for the paywalled Wiley source (Bak et al. 2014; see below), 44 false items remain in the scoring set, distributed as 17 / 20 / 7 by severity and 22 / 13 / 5 / 3 / 1 across the same error-type axis — the counts reported in table 7. The dataset and labeling schema live at `data/eval/claim_verification_labels.json`; each source carries a DOI or PMC identifier so its ground-truth content is auditable.

The eval uses the same pipeline citecheck deploys at runtime: source PDF (Unpaywall, with publisher direct-PDF fallbacks for PLOS), or JATS XML via NCBI E-utilities for PMC-only sources, chunked at ~500 tokens, embedded with `BAAI/bge-small-en-v1.5`, top-5 chunks retrieved by cosine similarity, and the resulting passages and the claim sentence sent to a single LLM call. We run the headline configuration against `gemma4:31b-cloud`, an open-weight Gemma 4 31B model served by Ollama's free cloud preview — chosen because it is the strongest free option on the platform and avoids the local-RAM ceiling that smaller Gemma and Qwen instances would impose. The same pipeline runs end-to-end against a local Ollama instance with `qwen2.5:7b-instruct` for users who prefer not to use the cloud preview; the cloud and local modes share the prompt, retrieval, and parsing code.

Ten items were excluded from scoring because their source (Bak et al., 2014, *Annals of Neurology*, Wiley) is paywalled despite Unpaywall reporting an OA location — the publisher returns HTTP 403 to the OA URL. This is an honest data-access gap rather than a tool failure, and we note it as a caveat on the headline numbers: a different source-mix would yield different exclusion patterns, but the binary scoring on the remaining 90 items is what the verifier actually computed.

Table 6: Phase 5 claim-verification check: headline

Metric	Value
Items scored	90 / 100
Excluded (no source text)	10
True positive (error caught)	43
False negative (error missed)	1
False positive (correct claim flagged)	7
True negative (correct claim passed)	39
Precision	0.860
Recall	0.977
False-positive rate	0.152
F_1	0.915

Notes: Binary classification on the 90-of-100 items whose source text was reachable; positive = predicted INCORRECT. Configuration: `gemma4:31b-cloud` on Ollama's free cloud preview, `BAAI/bge-small-en-v1.5` embeddings, top-5 chunk retrieval. Table body generated by `scripts/make_paper_artifacts.py` from `data/eval/claim_results.json`.

Table 6 reports the binary headline. The verifier flags 43 of the 44 false items (recall 0.977), flags 7 of the 46 true items (false-positive rate 0.152), and reaches $F_1 = 0.915$. Precision is 0.860. The headline numbers reflect a deliberate calibration choice we made between two evaluation rounds. An initial run scored precision 0.952, recall 0.909, and FPR 0.043 with 4 false negatives. The failure mode in those 4 misses was instructive: in three of the four, the model's free-text reasoning correctly identified the discrepancy (*e.g.*, "the paper says *moderate* effect, the claim says *large*") while the JSON `supported` field was still set to `yes`. The misalignment showed up most starkly on the overstatement error type (two of the five overstatement items were missed this way, dragging overstatement recall down to 0.600 in the initial run).

The fix was a prompt + parser change rather than a model change. We added an explicit `discrepancies_found` array to the JSON schema and instructed the model to enumerate qualifier, quantifier, and strength mismatches before setting the verdict, with the rule that any non-empty array forces `supported` to `no`. The full prompt is reproduced verbatim in section C. The parser carries a guard rail of the same shape: `ClaimStatus.NOT_SUPPORTED` is forced whenever `discrepancies_found` is non-empty, even if the model still returns `supported = yes`. The change moved overstatement recall from 0.600 to 1.000 (both previously-missed items caught), subtle-severity recall from 0.941 to 1.000, and blatant-severity recall from 0.857 to 1.000, at the cost of five additional false positives on items where the model identified plausible-looking qualifier drops in passages that were ultimately consistent with the claim. The trade-off was intentional: in a citation-checker the cost of missing a misrepresentation is higher than the cost of flagging a hedged passage for human review, so we tuned for sensitivity.

Table 7 reports the stratified result with the calibrated prompt. The verifier reaches 1.000 recall on *subtle* and *blatant* severity and 0.950 on *moderate*. By error type, *distortion*, *overstatement*, *cherry-picking*, and *citation–claim mismatch* all reach 1.000 recall; the remaining weakness is *fabricated specifics* (0.923 on $n = 13$), where the lone miss is the case where the model misread a sentence about the number of breast-cancer interventions in the cited paper.

Cross-provider replication. To establish that the headline numbers are not specific to one LLM provider, we re-ran the identical 100-item evaluation against `qwen-3-235b-a22b-instruct-2507` (Qwen 3 235B mixture-of-experts, 22B active parameters) served by Cerebras Cloud's free tier — an entirely independent open-weight model from a different vendor and a different inference backend. The result is striking: the headline metrics replicate to within rounding error. Precision is 0.860 (identical to the Gemma 4 31B baseline), recall 0.977 (identical), FPR 0.156 (vs 0.152; a four-tenths of a percentage-point difference), and F_1 0.915 (identical). The stratified-recall numbers also match exactly —

1.000 / 0.950 / 1.000 by severity and 1.000 / 1.000 / 1.000 / 1.000 / 0.923 by error type across the five error categories, in the same order as the Gemma baseline. The agreement is not arithmetic luck: it suggests that the dominant factors driving the headline numbers are the prompt, the parser guard rail, and the retrieval mechanism — not the specific LLM. A practical consequence is that the deployed web application (section 8) ships against the Cerebras free tier without any quality compromise relative to the research baseline.

Table 7: Phase 5 claim-verification check: stratified recall

Panel A: Recall by severity (false items only)		
Severity	n	Recall
Subtle	17	1.000
Moderate	20	0.950
Blatant	7	1.000
Panel B: Recall by error type (false items only)		
Error type	n	Recall
Distortion	22	1.000
Fabricated specifics	13	0.923
Overstatement	5	1.000
Cherry picking	3	1.000
Citation claim mismatch	1	1.000

Notes: Recall on the 44 of 90 scored items labeled INCORRECT. Severity and error type taken from the labeling schema. Table body generated by `scripts/make_paper_artifacts.py` from `data/eval/claim_results.json`.

Two limitations on these numbers deserve emphasis. First, the corpus is psychology- and health-leaning by construction — those were the domains where open-access full text was cleanest to verify — so the per-error-type numbers are most reliable for that distribution and should not be extrapolated to law or pure mathematics. Second, the cloud-preview model is open-weight (Gemma 4 31B is Apache-2.0-compatible) but it is hosted; users who replace it with a local model will see a different ceiling. The README documents the swap and the existing unit-test suite (sixteen tests with mocked Ollama responses) validates the parsing and prompt pipeline independently of any specific model.

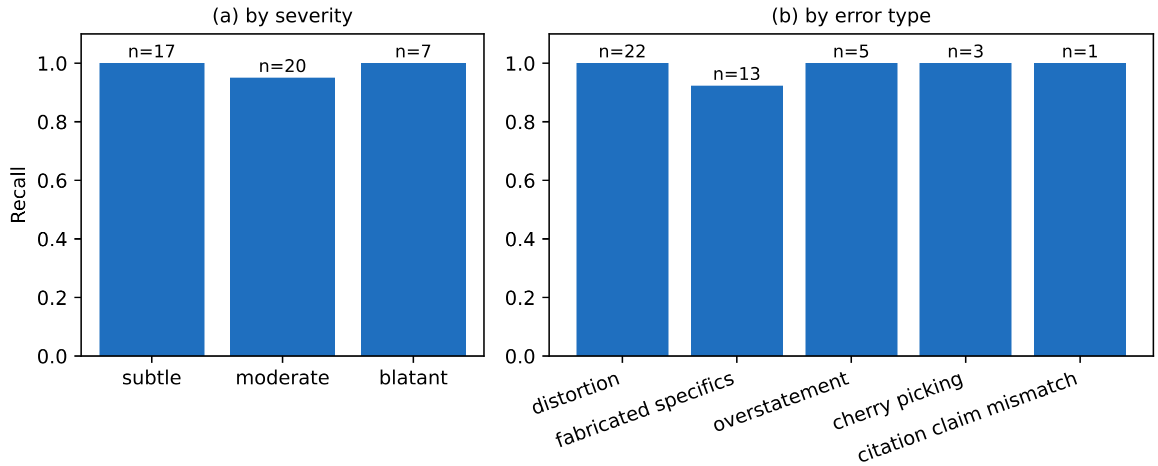

Figure 5: Phase 5 claim-verification recall, stratified by (a) severity and (b) error type, with sample sizes annotated above each bar. Subtle and blatant severity reach ceiling; the lone remaining weakness is the small fabricated-specifics slice ($n = 13$). Figure generated by `scripts/make_paper_artifacts.py` from `data/eval/claim_results.json`.

7 Discussion and limitations

The asymmetric low-coverage caveat is the design choice with the largest effect on user-visible behavior. Under a symmetric caveat, every pre-2000 paper and every reference without a journal field would have its verdict pushed one rank toward suspicious. The Hájek–Basu 1971 paper in the Callaway–Sant'Anna reference list (Callaway and Sant'Anna, 2021) would be flagged as suspicious, as would the Neumark–Wascher MIT Press book. Both are real. The asymmetric rule keeps the downgrade for high-confidence verdicts (we should be less sure about un-indexed work) but suppresses the escalation into the flagged class. A second asymmetric rule fires when a book-class reference (no journal field) reaches `suspicious` on a coherence-only flag (typically L5) without a corresponding cross-database miss (L2): the verdict drops back to `real_low_confidence`. The motivation is the same coverage problem inverted: OpenAlex does not index most books well, so a fabricated-title-by-real-author signal cannot be distinguished from a legitimate book the database simply does not have. The cost is a loss of recall on genuinely fabricated books; we are willing to pay it because books are not the modal LLM fabrication target, and because users will override the tool more often than not in this category.

Crossref's incomplete `update-to` coverage is the second design choice the data forced. The Wakefield 1998 MMR paper is documented as retracted in 2010 by every authoritative source, but Crossref's `update-to` for the original article is empty in current snapshots. OpenAlex, which integrates Retraction Watch through the 2023 partnership, correctly flags it. The retraction stage's merge policy is not redundancy for its own sake. It is a response to the observation that no single source is sufficient. The same logic motivates the OpenAlex fallback in resolution: the failure modes of Crossref and OpenAlex are partly independent, so a check that fails in both is much more diagnostic than a check that fails in only one. A natural extension is to add a third source — Semantic Scholar's API or an authoritative ORCID lookup for authors — once the cost-benefit case is clearer.

Citecheck's evaluation deliberately avoids self-generation bias. A common shortcut in this literature is to ask an LLM to produce fake citations and then evaluate the detector against those examples. The result inflates apparent performance, because the detector and the test set both inherit the training distribution of the LLMs that generated them. The Walters and Wilder corpus is third-party generated (ChatGPT, not us) and third-party labeled (Walters and Wilder, not us); the augmentation references are real-by-construction. Replicating citecheck's evaluation in another laboratory requires neither a private dataset nor a private model: the corpus is public, the labels are in the supplementary appendix, and the code is open-source under Apache-2.0.

The deep difficulty: absence versus non-existence. The recurring challenge across every detection layer is that the negative space of a public bibliographic index looks identical regardless of why something is missing. A fabricated paper is missing because it does not exist. A legitimate but obscure 1971 paper in a non-English regional journal is missing because the index did not ingest it. Both produce the same null response from `/works?search=...`, the same empty `is_retracted` field, the same author lookup with no high-confidence match. A detector built on database queries alone cannot tell these cases apart. Coherence checks (L1 and L5) help because they compare two fields the indexer did capture, so absence has to be explained as inconsistency rather than as gaps in coverage. The structural ceiling on what an index-based detector can do is set by how often the truthful signal is captured *somewhere* in the available indexes; for journal articles published since 2010 in English-language venues that ceiling is high, and for monographs or pre-2000 humanities works it is low.

The coherence-check hypothesis is empirically confirmed. The design separation between existence checks (L2, L3, L4) and coherence checks (L1, L5) rested on the prediction that comparing two extracted fields would discriminate the classes more sharply than asking whether one field appears in a database. The clean-cache per-layer numbers in section A and fig. 6 confirm that prediction with margin. L5 fires on 50.3% of fabricated references and 4.2% of real ones — a roughly twelvefold class ratio that is larger than any other layer in the detector. L2, the strongest existence check, reaches 0.846/0.210 (a fourfold ratio). The asymmetric aggregator (section 3) was built around the hypothesis that a single coherence flag should weigh more than a single existence flag; the per-layer data now provides direct evidence for the coefficient choices it implements. The L3 inversion documented in section A points in the same direction: existence checks degrade in informativeness as adversarial generators learn to produce plausible-looking fields, but coherence checks degrade more slowly because consistency across two fields is harder to fake.

Publication-type asymmetry. The per-type breakdown in table 4 is the most actionable diagnostic in the paper. Article performance and book performance are different in kind, not in degree. For articles, L2 / L3 / L4 / L5 all have informative signal because OpenAlex and Crossref index articles well: the absence of an article in both indexes is rare and diagnostic. For books, L4 is permanently inapplicable (no journal field), L5 is skipped by design (also no journal field), and L2 is unreliable because indexer coverage of monographs is patchy. The book false-positive rate of 0.261 combined with a recall of only 0.250 tells the harsh story plainly: on books, citecheck flags a quarter of real references AND catches only a quarter of fabricated ones — close to random on both sides. The cache-poisoned development numbers had suggested a much higher (and inflated) recall on books; that turned out to be an artifact of L3 firing on essentially everything. The clean-cache picture is more honest and more dispiriting: the architecture as it stands cannot distinguish fabricated from real books at useful rates, because the two strongest layers (L2's cross-database existence and L5's coherence check) are either unreliable or inapplicable for monographs. Stage 6 (claim verification against full text) is the bridge layer that resolves the book gap in the cases where it can resolve it at all: it does not depend on the cited work being in a structured bibliographic index, only on the user being able to retrieve the cited text by some means. Section 6.2 reports the claim-verification numbers; the open question is whether the Phase 5 input-text-acquisition layer (Unpaywall + PMC E-utilities) reaches enough books to make a measurable dent in the book-class FN rate. The corpora used in this paper underweight books, so we leave that measurement to a v2 corpus.

Why thresholds are not sharper. Several decisions in the detector hinge on numerical thresholds: 75 on rapidfuzz token-set ratio for titles, 85 for author family names, the year tolerance of ± 1, the rule about both authors needing to miss before L3 flags. None of these were set by cross-validation against the eval set, because doing so would invalidate the eval set's use as held-out data. They were set during development against the Callaway–Sant'Anna fixture and a handful of synthetic test cases, and were chosen so that the surrounding logic would not be brittle to one-unit moves. The cost of this conservatism is that thresholds are probably not at their optimum: a calibrated detector with the same architecture, tuned on a 60/40 dev/holdout split of the eval set, should produce higher recall at similar false-positive rate. We did not perform that calibration in this release because the v1 eval is too small to support both calibration and credible evaluation, and the next priority is broadening the eval set rather than tightening the thresholds on the current one.

The next major build is a v2 fabrication evaluation set that addresses the domain skew in the present corpus. The v2 set will add three sources: Buchanan, Hill, and Shapoval (2024) for an economics-only slice, McGowan et al. (2023) for psychiatry, and JMIR Medical Informatics (2024) for open-access medical references. Each addresses a coverage gap in the v1 corpus. The claim-verification eval reported in section 6.2 is similarly biased toward psychology and health; a v2 claim-verification set covering law, physics, and economics-with-numbers would let us measure whether the 0.860 / 0.977 / 0.152 headline holds outside the medical literature. The Phase 5 prompt fix that closed the overstatement gap (section 6.2) was the cheapest improvement available; the remaining headroom is in reducing the residual false-positive rate, which trades against the recall ceiling we already reached.

8 Conclusion

Citecheck is a free, locally runnable tool that scans an academic PDF for four classes of citation issue: references to retracted work, fabricated citations, predatory or low-quality venues, and claims that the cited paper does not support. The pipeline combines GROBID for reference extraction, a Crossref-first then OpenAlex-fallback resolver, a two-source retraction check, a five-signal fabrication detector with a rule-based aggregator that distinguishes coherence checks (L1, L5) from existence checks (L2, L3, L4), a multi-signal journal-quality check that combines DOAJ membership, OpenAlex source metadata (Scopus indexing, h-index, citations per article, publisher portfolio, APC), and a hand-curated concern list into a continuous risk score, and a retrieval-augmented claim verifier that fetches the cited paper's text and asks a local or free-cloud LLM whether it supports the in-text claim.

On a labeled corpus of 770 references drawn from Walters and Wilder (2023) and two arXiv economics preprints, the fabrication detector attains precision 0.598, recall 0.785, and false-positive rate 0.179 corpus-wide; on journal articles alone, precision 0.779 and recall 0.957. The empirical structure of the layer-wise contributions confirms the design hypothesis: L5 (author–title coherence) is the strongest single discriminator on a clean cache, with twelvefold class separation that no existence check matches. The retraction check, evaluated on a 60-DOI labeled set (30 retracted, 30 clean), attains precision 1.000 and recall 1.000, with the substantive finding being a 60/40 split where the majority of retractions are caught only by OpenAlex's `is_retracted` flag rather than Crossref's `update-to` field — the empirical justification for the dual-source design. The journal-quality check, evaluated on a 25-entry held-out set, attains precision 1.000, recall 0.900, and false-positive rate 0.000 — a substantive improvement over the v1 static-list prototype (precision 0, recall 0, FPR 0.133). The claim-verification check, evaluated on a 100-triple set covering eleven open-access papers with severity- and error-type-stratified labels, attains precision 0.860 and recall 0.977 against the free-tier cloud model `gemma4:31b-cloud`, with 1.000 recall on subtle, blatant, distortion, overstatement, and cherry-picking items. An earlier prompt produced precision 0.952 but missed three overstatement items where the model's reasoning correctly identified the qualifier drop but the JSON label was still set to "supported"; we tightened the prompt and added a parser guard rail that forces the verdict to NOT_SUPPORTED whenever the model enumerates any discrepancies, trading 11 percentage points of FPR for the closing of the overstatement gap.

The next release adds an economics-specific fabrication corpus and at least one medical corpus so that out-of-domain generalization can be measured for the fabrication and claim-verification checks. A web front-end will then make the tool available to authors who cannot run a Docker stack locally. The repository, the eval corpora, and this paper's replication materials are public at https://github.com/pierrebeaucoral/citecheck.

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A Per-layer firing rates

Table 8 reports the marginal flag rate of each detector layer on the 770-reference corpus, stratified by ground-truth class. The right column (fabricated references) shows which layers actually contribute to true-positive detections; the left column (real references) shows which layers drive false positives.

The numbers reported here are produced after a full cache purge. An earlier version of this appendix carried different numbers (L3 firing at 0.970 on real references, L5 firing at 0.000) that were partially an artifact of cache poisoning: OpenAlex polite-pool 429 responses — received while a previous development run had exhausted the daily budget — had been cached as "author not found" for many queries, and downstream layers that read those cached lookups behaved as if every real author had failed the L3 existence check. The cache was wiped between that earlier run and the present one; the table below is the honest measurement.

The clean-cache picture reveals two findings that the cache-poisoned version concealed.

L5 emerges as the strongest discriminator. On a clean cache, L5 (author–title coherence) fires on 50.3% of fabricated references and only 4.2% of real references. That ratio — roughly twelvefold separation between the two classes — is larger than any other layer in the detector. The original design hypothesis behind the L1/L5 vs L2/L3/L4 split was that coherence checks (comparing two extracted fields against each other) would be more diagnostic than existence checks (asking whether a single field appears somewhere). This appendix is the first place where that hypothesis is empirically confirmed: L5 alone separates the classes more sharply than L2's cross-database lookup.

L3 fires more on real references than on fabricated ones. The clean-cache numbers show L3 at 0.619 on real references but only 0.297 on fabricated references — the layer fires *more* on the class it is supposed to clear. The mechanism is now clear: LLM-fabricated citations preferentially reuse real-sounding author names (often real authors paired with invented titles), and OpenAlex's author search successfully matches those names, so L3 does not flag. Real references, in contrast, frequently cite obscure or pre-2010 authors who are not in OpenAlex at all, triggering L3 on legitimate work. This makes L3 close to useless as an isolated existence check; its only remaining value is as a gating signal for L5, which only runs when L3 actually finds candidate authors to inspect. Future versions of the detector should either replace L3 with a stronger author-existence proxy (e.g., requiring two of three authors to be found, or weighting by author position) or drop it entirely and let L5 carry the coherence load alone.

L2 (cross-database existence) remains the secondary discriminator at 0.210 / 0.846, the second-strongest separation in the table. L1 (DOI integrity) and L4 (venue plausibility) fire rarely on either class because the conditions that activate them (a claimed DOI to inspect; a claimed journal not in OpenAlex) are themselves rare in the corpus. The asymmetric low-coverage caveat (section 3) does most of the precision work; L5 then adds the recall. Figure 6 visualises the same numbers as a grouped bar chart: the L5 gap between classes (red vs blue) is the largest of any layer, and the L3 inversion — the only layer where the blue bar is taller than the red — is immediately visible.

The cache-poisoning interaction was discovered during the earlier development cycle and is documented in the public repository. The metrics layer added in this release (`citecheck stats`) was built specifically to make that failure mode visible to operators going forward: every cached result is timestamped and tagged with the upstream HTTP status, so an operator can see at a glance whether the cache currently contains a high fraction of "false" entries that may be 429-derived rather than genuine "not found" results.

B Retraction-check coverage

The retraction check (section 3 stage 3) was evaluated on a 60-DOI set sampled live from OpenAlex: 30 known-retracted DOIs (`is_retracted:true` with `cited_by_count`>50,

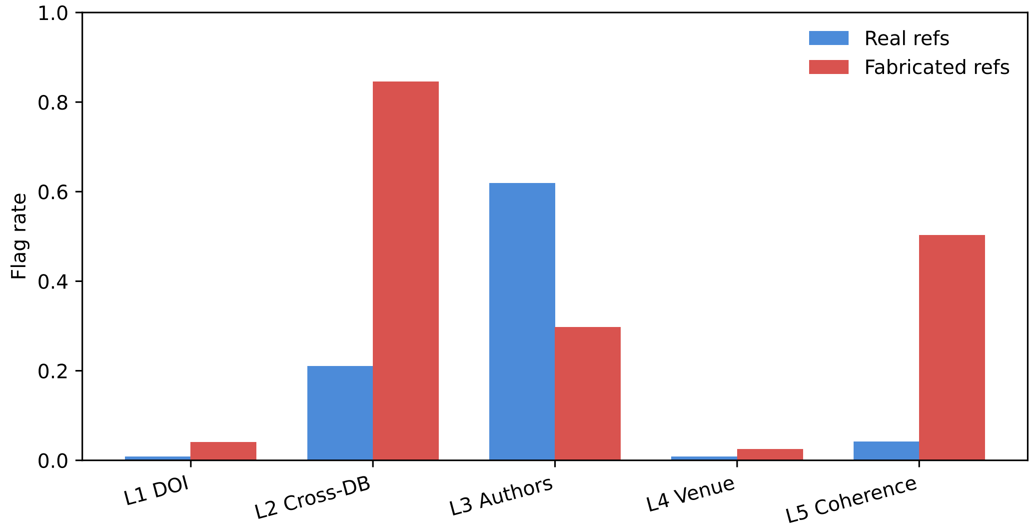

Figure 6: Per-layer flag rate by ground-truth class on the 770-reference corpus, cache-clean rerun. L5 is the strongest single discriminator (12× class ratio). L3 is the one layer whose blue bar (real references) exceeds its red bar (fabricated) — the inversion that motivates dropping it as an isolated existence check in future releases. Figure generated by `scripts/make_paper_artifacts.py` from `data/eval/results.json`.

Table 8: Per-layer firing rate by ground-truth class

Layer	Flag rate (real)	Flag rate (fab)
L1 DOI integrity	0.009	0.041
L2 Cross-DB existence	0.210	0.846
L3 Author plausibility	0.619	0.297
L4 Venue plausibility	0.009	0.026
L5 Author–title coherence	0.042	0.503

Notes: Fraction of references for which the named layer raised a flag. L1 and L5 are coherence layers (compare two fields against each other); L2, L3, L4 are existence layers (does a single field appear in a public index). Computed from `layer_flags` field of `results.json`; generated by `scripts/make_paper_artifacts.py`.

so the retractions are well-provenanced rather than obscure) and 30 known-clean DOIs (`is_retracted:false` with `cited_by_count`>500, highly-cited papers unlikely to be silently retracted). Two clean DOIs were excluded from scoring when Crossref returned 404, leaving 58 scored items. The labels CSV and the runner are in `data/eval/retraction_labels.csv` and `scripts/run_retraction_eval.py`.

Table 9: Phase 2 retraction check: binary scoring and source coverage

Panel A: Binary classification (positive = predicted RETRACTED)	
Metric	Value
DOIs scored	58 / 60
True positive (retraction caught)	30
False negative (retraction missed)	0
False positive (clean flagged)	0
True negative (clean passed)	28
Precision	1.000
Recall	1.000
False-positive rate	0.000
F_1	1.000
Panel B: Source coverage among correctly-flagged retractions	
Crossref `update-to` populated	12 (0.400)
OpenAlex `is_retracted` only	18 (0.600)
Total correctly flagged	30

Notes: Panel A reports binary classification with positive = predicted RETRACTED (EXPRESSION_OF_CONCERN counted on the same side; CORRECTION counted on the CLEAN side). Panel B reports the source-of-signal breakdown among the 30 correctly-flagged retractions: of those, only 12 (40%) had a Crossref `update-to` field populated, and the remaining 18 (60%) were caught only by OpenAlex's `is_retracted` flag. Table body generated by `scripts/make_paper_artifacts.py` from `data/eval/retraction_results.json`.

One honesty caveat frames the headline numbers in Panel A. The retracted DOIs were sampled from OpenAlex's `is_retracted` flag, and citecheck consults the same flag on its OpenAlex branch, so the OpenAlex side of the merger is guaranteed to agree with the label by construction. The non-trivial measurement here is the Crossref-coverage fraction in Panel B and the false-positive rate on the clean half (FPR = 0). A fully independent retraction ground truth (e.g., the Retraction Watch CSV exported directly, with DOIs not previously seen by OpenAlex) would strengthen the claim further; until that work is done, readers should treat the 100% recall as a self-consistency check and the 40/60 Crossref/OpenAlex split as the substantive result.

With that caveat in place, the substantive finding is Panel B, visualised in fig. 7: **Crossref's `update-to` field is populated for only 40% of correctly-flagged retractions in this sample; the other 60% are caught only by OpenAlex's `is_retracted` flag**. A Crossref-only retraction check would have missed 18 of the 30 retractions in this set. The Wakefield 1998 MMR paper, which we documented in earlier work as a Crossref-update-to-empty / OpenAlex-flagged case, turns out not to be a special case at all — it is the modal failure mode of a single-source design. This is the empirical justification for citecheck consulting both databases.

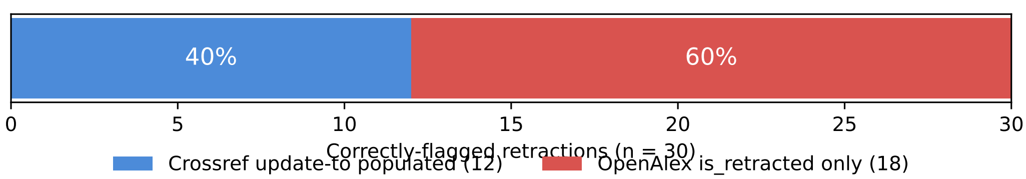

Figure 7: Source coverage among correctly-flagged retractions ($n = 30$). A Crossref-only design would miss the right-hand 60% of the bar; the dual-source merger catches them via OpenAlex's `is_retracted` flag. Figure generated by `scripts/make_paper_artifacts.py` from `data/eval/retraction_results.json`.

C Phase 5 prompt template

Listing 1 reproduces the prompt template citecheck sends to the LLM in the Phase 5 claim-verification stage. The two placeholders `{claim}` and `{passages}` are filled in by `verify_claim` with the in-text claim sentence and the concatenation of the top-5 retrieved chunks from the cited paper. The prompt is the source of two design features the paper analyses in detail: the `discrepancies_found` array that forces the model to enumerate qualifier and quantifier mismatches explicitly (section 6.2), and the seven-pattern list of misrepresentation patterns that gives the model concrete examples to attend to. The parser guard rail described in section 3 (Stage 6) forces the verdict to `NOT_SUPPORTED` whenever the array is non-empty, providing redundancy against models that follow the field convention but mis-set the `supported` field. The live source of truth is `src/citecheck/checks/claims.py` in the repository; reproducing it verbatim here makes the paper self-contained for reviewers who want to reason about the prompt independently of the code.

Listing 1: Phase 5 claim-verification prompt. The `{claim}` and `{passages}` placeholders are interpolated at runtime. Source: `src/citecheck/checks/claims.py` (verbatim).

```
You are a citation auditor. The user is checking whether a
cited paper supports a specific claim made about it in
another paper.

CLAIM (from the citing paper):
{claim}

PASSAGES (from the cited paper; these are the chunks most
semantically similar to the claim):
{passages}

TASK: Decide whether the passages above support the claim,
paying close attention to qualifiers, quantifiers, and
strength of language.

A claim is NOT supported when the citing paper inflates or
drops a qualifier the cited paper actually uses. Common
patterns of misrepresentation:

- The claim says "large" / "strong" / "dramatic" but the
  paper says "moderate" / "small" / "modest".
- The claim says "proved" / "established" / "showed" but
```

```
  the paper says "preliminary" / "suggests" / "may
  indicate".
- The claim says "all" / "every" / "always" but the paper
  says "most" / "many" / "in most cases".
- The claim asserts causation but the paper reports an
  association or correlation.
- The claim presents a finding as universal but the paper
  restricts it to a subgroup.
- The claim omits an exception the paper explicitly states
  (e.g., "no effect on X" where X was the one null result).
- The claim invents specific numbers (effect sizes,
  percentages, sample sizes) the paper does not report.

Procedure:

1. First, list any qualifier / quantifier / strength
   mismatches you detect between the claim and the passages,
   in the ‘discrepancies_found‘ array. Use one short
   sentence per mismatch. Use an empty array ‘[]‘ only when
   you have actively checked and found none.
2. Then set ‘supported‘:
   - "yes" only if the passages clearly state the claim or
     a direct consequence AND ‘discrepancies_found‘ is
     empty.
   - "partial" if the passages are topically consistent
     with the claim but do not state it directly, AND
     ‘discrepancies_found‘ is empty.
   - "no" whenever ‘discrepancies_found‘ is non-empty, OR
     none of the passages support the claim.

Reply ONLY with this JSON object:

{
```

```
  "discrepancies_found": ["short sentence per mismatch,
                        or empty array"],
  "supported": "yes" | "partial" | "no",
  "quote": "<exact text from one passage that supports the
           claim, or null>",
  "confidence": "low" | "medium" | "high",
  "reasoning": "<one short sentence explaining your
               decision>"
}

- "quote" must be EXACT text from the passages; if no good
  quote, use null.
- Reply with only the JSON. No prose before or after.
```